

SRS VERSION 1.0

DRAFT

GymX – Smart Gym Management System

An enterprise-level web application designed to automate and simplify the daily operations of fitness centres, health clubs, and sports facilities. GymX centralises all business operations into a single, secure platform, enabling administrators, employees, coaches, nutritionists, accountants, and members to manage memberships, workouts, attendance, payments, reports, and communication with ease.

Built on a modern technology stack — **Django**, **Django REST Framework**, **React.js / Next.js**, and **PostgreSQL** — the system follows a modular architecture that supports scalability, maintainability, and seamless future integrations with mobile applications and third-party services.

Prepared by Mostafa Mohamed Mahmoud — June 2026

Sections 1–3

Introduction, Purpose & Scope

GymX replaces traditional, manual gym management processes with a comprehensive digital ecosystem. The platform enables gym owners and staff to manage members, coaches, employees, attendance, memberships, classes, workout plans, nutrition plans, payments, reports, inventory, equipment, and multiple branches — all through one centralised dashboard. Secure authentication, role-based access control, analytics, and RESTful APIs ensure the system is ready for future mobile application development.

Introduction

GymX is a modern, enterprise-level Gym Management System designed to automate and simplify fitness centre operations. It centralises all business functions into a single, secure web application with a modular, scalable architecture.

Purpose

This SRS defines all functional and non-functional requirements. It serves as the primary reference for developers, testers, project managers, UI/UX designers, and stakeholders throughout the software development lifecycle.

Scope

The platform delivers a complete digital ecosystem: memberships, classes, workout and nutrition plans, payments, reporting, inventory, equipment, and multi-branch management — all accessible via RESTful APIs.

Sections 4–5

Definitions & References

This document employs a range of technical acronyms and industry standards. Familiarity with these terms is essential for all team members involved in implementation, testing, and ongoing maintenance. The references listed below provide the foundational guidelines that govern GymX's architectural and security decisions.

Key Terms & Acronyms

Term	Meaning
SRS	Software Requirements Specification
API	Application Programming Interface
JWT	JSON Web Token
RBAC	Role-Based Access Control
CRUD	Create, Read, Update, Delete
QR	Quick Response Code

Standards & Technologies Referenced

- IEEE 29148 Software Requirements Specification
- Django & Django REST Framework Documentation
- PostgreSQL Database Documentation
- React Frontend Documentation
- JWT Authentication Standards
- REST API Best Practices
- OWASP Security Guidelines

Section 6

Product Overview

GymX is a centralised web platform that replaces traditional, manual gym management processes with an automated digital solution. It allows gym owners and staff to efficiently manage all aspects of daily operations — from membership tracking and class scheduling to financial reporting and equipment inventory — whilst simultaneously improving the member experience and increasing overall business productivity.

The system is designed to support multiple user roles, each with tailored permissions and dashboards. Advanced reporting and real-time analytics provide actionable business intelligence, whilst a scalable, secure architecture ensures the platform can grow alongside the business and integrate with future mobile and IoT extensions.

- ❏ GymX consolidates memberships, classes, payments, reporting, inventory, and multi-branch management into a single, secure, role-aware dashboard — eliminating manual workflows and enabling data-driven decision-making.

Section 7

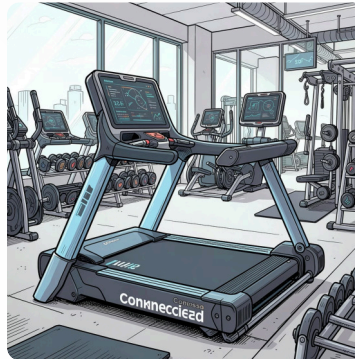
Product Perspective & Future Integrations

GymX is developed as an independent enterprise software solution following a modular architecture. Each business function — whether membership management, class scheduling, payment processing, or inventory tracking — is implemented as an independent module while maintaining seamless integration across the entire system. This approach significantly improves maintainability, scalability, testing efficiency, and the capacity for future expansion without disrupting existing functionality.



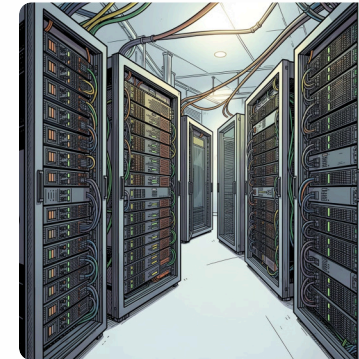
Mobile & Wearables

Native mobile applications and smart watch integrations to extend the platform's reach beyond the desktop, providing members and coaches with on-the-go access to schedules, plans, and attendance tracking.



IoT & AI Services

Direct integration with IoT-enabled fitness equipment and AI-driven recommendation services to deliver personalised workout and nutrition insights based on real-time performance data.



Cloud, Payments & APIs

Scalable cloud storage, third-party online payment gateways, and external API integrations to support business expansion, automated billing, and seamless data exchange with partner services.

Sections 8–9

Product Objectives & Business Goals

GymX is driven by a clear set of product objectives and measurable business goals. The platform aims to automate gym operations, improve operational efficiency, and reduce administrative paperwork — all whilst elevating the member experience. From a business standpoint, GymX targets increased customer satisfaction, improved financial management, and support for multi-branch expansion through accurate, real-time reporting and modern digital tools.



Section 10

Stakeholders & User Roles

A diverse group of stakeholders interacts with GymX, each with carefully defined roles and permissions enforced through Role-Based Access Control (RBAC). From gym owners and system administrators to coaches, nutritionists, and members — every user type accesses a tailored interface reflecting their responsibilities and authority level within the platform.



Management

Gym Owners, Business Managers, System Administrators, and Branch Managers responsible for strategic, operational, and administrative oversight across the organisation.



Members

Gym members who interact with the platform to view schedules, track attendance, access workout and nutrition plans, manage payments, and communicate with staff.



Staff

Receptionists, Coaches, Nutritionists, Accountants, and Maintenance Staff executing day-to-day gym operations, member services, financial tracking, and equipment upkeep.



Technical

Software Development Team and Technical Support Team responsible for building, maintaining, securing, and extending the platform throughout its lifecycle.

Section 11

Operating Environment

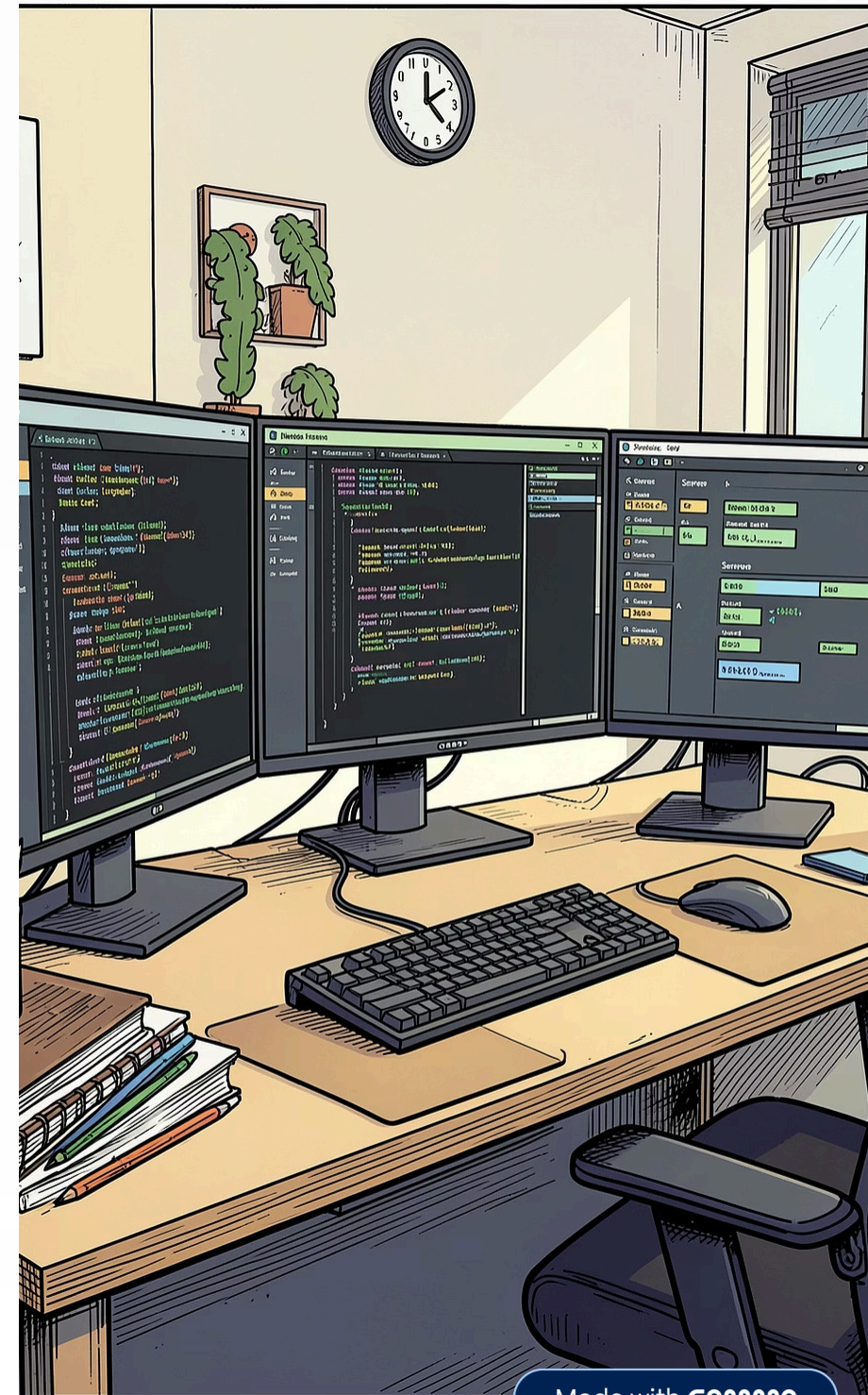
GymX is designed to operate across a broad range of modern web browsers and operating systems, ensuring accessibility for all user roles. The backend infrastructure leverages containerised deployments for consistency, portability, and scalability, with a robust reverse proxy and a high-performance relational database at its core.

Client Side

Browsers: Chrome, Edge, Firefox, Safari
OS: Windows, macOS, Linux, Android, iOS
Access: Responsive web application

Server Side

OS: Linux servers
Containers: Docker
Web Server: Nginx
Database: PostgreSQL



Sections 12–13

Assumptions & Constraints

Before and during development, several fundamental assumptions are made regarding user infrastructure and available services. Equally, a set of technical and architectural constraints define the boundaries within which GymX must be built, ensuring security, consistency, and long-term scalability.

Assumptions

- Users have reliable internet access.
- Staff receive basic system training.
- Email services are available for notifications.
- Payment gateway services are operational.
- Stable server infrastructure is maintained.
- Members possess valid contact information.

Constraints

- **Framework:** Django + Django REST Framework
- **Authentication:** JWT-based token authentication
- **Database:** PostgreSQL
- **Communication:** Secure HTTPS only
- **Access Control:** RBAC enforced throughout
- **Scalability:** Architecture must support future scaling without major redesign

Document Control & Summary

GymX is positioned to be a transformative, enterprise-grade platform for the fitness industry. By replacing fragmented, manual processes with a unified, modular, and secure digital ecosystem, GymX delivers tangible value across every layer of gym operations — from front-desk check-ins and class scheduling to financial reporting and multi-branch oversight.

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User Roles

Owners, Admins, Staff, Coaches, Members, Support

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Future Integrations

Mobile, Wearables, IoT, AI, Payments, Cloud, APIs

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Tech Stack Layers

Django, DRF, React, PostgreSQL, Docker

1.0

SRS Version

Initial Draft, June 2026

This Software Requirements Specification serves as the authoritative reference for the GymX development team. All functional and non-functional requirements outlined herein are to be verified and validated through structured testing prior to release. Stakeholders should direct feedback to the project lead for inclusion in subsequent revisions.